

National Research University Higher School of Economics

St. Petersburg Branch

St. Petersburg School of Economics and Management

**Econometrics Project: “The impact of countries’
socio-economic attractiveness on foreign direct
investments inflows”**

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Confirmation of the originality of the econometric's project

I, Dumanov M. M., BMB2209, 3rd year student of International Bachelor in Business and Economics Programme, HSE Saint Petersburg School of Economics and Management confirm, that econometric's project was

made by myself and:

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2. does not reproduce the work done by other authors, without indicating a link to the source of educational or scientific literature, articles, websites, completed assignments or notes of other students;
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5. includes a complete bibliographic list of references and sources that have been used in writing the text of the report on the course project.

I am aware that violation of the citation and linking rules is considered as deception or an attempt to mislead, and also qualifies as a violation of the HSE Internal Regulations.

Confirmation of the usage of AI-technologies

I, Dumanov M. M., BMB2209, 3rd year students of International Bachelor in Business and Economics Programme, HSE Saint Petersburg School of Economics and Management confirm, that econometric's project

made using Artificial intelligence technologies in the following:

1. Artificial intelligence as ChatGPT-4 and analogues were used to translate a large amount of texts into English literally and succinctly;
2. DeepL translator based on AI for translating small texts and articles from Russian into English in a more literal way;
3. Automating the process of linking to sources in APA style

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Introduction

Relevance of the problem:

Foreign direct investment (FDI), which delivers capital, technology, experience, and jobs, significantly aids economic development. But the amount of FDI that is drawn in differs substantially from one nation to another, which emphasises how crucial it is to research the factors that affect these variations. One of the primary determinants of the nation's appeal to foreign investors is the ease of conducting business, which is a reflection of the degree of the regulatory framework, the accessibility of administrative processes, and the efficiency of institutions to safeguard the interests of entrepreneurs.

Ensuring ease of doing business is essential for drawing in foreign direct investment (FDI) since it lowers investment risks, increases investor trust, and fosters market competition. The degree of urbanisation, which affects market density and infrastructure availability, political stability, which reduces investor uncertainty, and population income (GDP per capita) are other factors that have a big impact on investment volumes.

It is crucial to examine how shifts in a nation's business-friendly environment affect FDI, especially over a given period of time (like 2019). First of all, it offers more profound understanding of the elements influencing investment choices and the processes underlying them within the framework of more general macroeconomic circumstances. Second, by investigating the relationship between ease of doing business and variables like GDP per capita, urbanisation, and political stability, we may identify intricate interdependencies and provide a comprehensive picture of a nation's appeal to investors.

The study's potential to provide well-founded suggestions for raising countries' competitiveness in the international investment arena is what gives it practical significance. This is particularly important for policymakers who want to create a stable atmosphere that draws in foreign capital and promotes long-term, sustainable economic growth.

Research Question:

How does the countries' level of ease of doing business affect the volume of attracted foreign direct investment in 2019?

Research Goal:

The goal is to study the relationship between the level of ease of doing business and the volume of foreign direct investment attracted by countries in 2019.

This study will illustrate how the ease of doing business and related macroeconomic and social parameters affect the attraction of foreign investment. The study's findings will provide us a better understanding of the processes that contribute to a nation's appeal to investors and

provide useful suggestions for creating policies that support economic expansion and advancement.

To achieve this objective, the following objectives were set:

- 1) Collect and analyze statistical data on the level of ease of doing business in countries in 2019
- 2) To collect and analyze statistical data on the volume of foreign direct investment (FDI) in countries during the same period.
- 3) To assess the impact of the level of ease of doing business on the amount of FDI attracted using statistical analysis techniques.
- 4) Determining the impact of GDP per capita, political stability, unemployment rate, corruption control and urbanization rate on the investment attractiveness of countries.
- 5) Analysis of the relationship between various factors (ease of doing business, GDP per capita, political stability, urbanization) and the volume of attracted foreign direct investment, taking into account the characteristics of countries with different levels of development.
- 6) Critical analysis of the obtained results and search for possible solutions to improve the model and optimize the study

Literature review

This literature review examines the impact of economic freedom on attracting foreign direct investment by analysing results from the following articles: "Economic freedom, Fiscal rules on FDI inflows: An analysis of 24 developing countries." by Mitsi, D. (2023), "Factors Influencing FDI Inflows in SouthAsian Countries: A Panel Data Analysis" by Md. Jobaer Hossain (2019), "Nexus between Ease of Doing Business and Foreign Direct Investment: Evidence from 130 Economies." Xu, X., Hu, Y., & Tahir, S. H. (2023) and "Investment attractiveness of the country: analyzing the main factors." by Ani Grigorievna Avetisyan (2020). The purpose of this review is to provide an overview of the literature, to emphasise the relevance of the topic, and to compare research findings.

Mitsi, D. (2023). Economic freedom, Fiscal rules on FDI inflows: An analysis of 24 developing countries.

This paper aims to explore the relationship between economic freedom, a new type of fiscal institution named fiscal rules, and FDI inflows. It provides a comparative analysis of different countries and investigates the mechanisms through which economic freedom and fiscal institutions influence FDI.

The panel data analysis employed in this study utilizes two estimation methods, namely the Random Effects Model (determined through the Hausman test) and the Two Stage Least Squares Method (to address endogeneity concerns). The empirical findings reveal several noteworthy insights. Firstly, GDP, trade openness, and gross fixed capital formation exhibit a positive relationship with FDI inflows, while inflation does not have a significant impact. Additionally, our research indicates that specific economic freedom sub-indicators, such as tax burden, monetary freedom, trade freedom, and financial freedom, positively influence FDI inflows. Conversely, the presence of expenditure rules is found to have a negative impact on FDI inflows. Furthermore, we explore the interactive effects of fiscal rules and economic sub-indicators on FDI inflows, providing further insights into the relationship between these factors.

Md. Jobaer Hossain (2019). "Factors Influencing FDI Inflows in SouthAsian Countries: A Panel Data Analysis" .

The study analyzes how various factors affect FDI inflows to South Asian countries. The authors examine data on FDI and economic performance from 1972 to 2016 and identify that market size (GDP) plays a key role in attracting FDI. This study can be useful in understanding the specific impact of economic conditions on investment inflows in developing countries

Xu, X., Hu, Y., & Tahir, S. H. (2023). Nexus between Ease of Doing Business and Foreign Direct Investment: Evidence from 130 Economies.

Based on both yearly data of 130 economies, this paper studies the association between the ease of doing business and FDI from three characteristics, i.e., internationalization, legalization, and facilitation. The results demonstrate that the ease of doing business and its three characteristics, i.e., internationalization, legalization, and facilitation all have a statistical positive influence in attracting FDI inflows. Second, this paper finds that in the legalization of doing business, the resolving insolvency indicator is the most statistically positive on attracting FDI, and in the facilitation of doing business, the paying taxes indicator plays an essential role in attracting FDI. Third, this paper finds that the ease of doing business has a vital impact on FDI inflows in developing countries, but is not significant for developed countries. The implications of this paper are helpful for local governments optimizing their business environment to attract FDI.

Ani Grigorievna Avetisyan (2020). Investment attractiveness of the country: analyzing the main factors.

Attracting investment is relevant for any economy. The purpose of this study is to examine the factors affecting the inflow of foreign direct investment using the example of countries leading in this indicator and the countries of the Eurasian Economic Union (EAEU). The author uses both qualitative and quantitative methods of analysis of research data of other authors, reports of international organizations and the World Bank database. . Linear regression model estimated by the least squares method is constructed. The basis for the calculations was the combined model of panel data analysis applied to the BRICS and MINT countries, which allowed the authors to calculate the degree of influence and significance of a number of variables on the volume of FDI inflows. It is shown that the factor of political stability is more significant than inflation, foreign trade and final consumption expenditures. The author concluded that the inefficiency of economic and political regulatory institutions is a constraint to investment growth. In particular, non-transparent regulatory policies, dominance of state ownership and lack of adequate investor protection, poor rule of law and restriction of economic freedoms are negative institutional factors. Further study of issues related to FDI attraction should take into account the trends of mass digitalization, as the introduction and diffusion of digital technologies is becoming an integral component of the competitiveness of modern economies.

Comment on the Literature Review:

This literature review presents a comprehensive study of the impact of economic freedom and institutional factors on attracting foreign direct investment (FDI). Mitsey (2023) emphasizes

the role of economic freedom and fiscal rules, highlighting their subtle impact on FDI, such as the positive impact of sub-indicators such as tax and fiscal freedom, along with the negative impact of spending constraints. Xu et al. (2023) highlight the importance of ease of doing business, especially in developing countries, emphasizing the significance of institutional reforms. Finally, Avetisyan (2020) focuses on political stability and institutional effectiveness, highlighting the crucial role of digitalization in enhancing economic competitiveness. The study by Md. Jobaer Hossainl (2020) complements the review by analyzing the role of institutional quality and market size (GDP), especially in South Asia, over the period 1972–2016, illustrating the importance of economic conditions in attracting FDI. This review not only organizes existing research but also highlights the need for further examination of the complex interaction between economic freedom, institutional factors, and FDI flows, offering a solid basis for policies to improve the investment climate.

Descriptive statistics, intelligent analysis of data

Since 2019 is the best time to examine how economic freedom and associated characteristics affect the ability to draw in foreign direct investment (FDI), data from that year were used in this study. The choice of 2019 is due to the fact that it predates the commencement of the COVID-19 pandemic, which has caused enormous distortions into global economic processes, including drastic shifts in FDI quantities, political stability and economic activity. By examining the 2019 data, it is possible to mitigate the effects of these anomalies and obtain more reliable results. Additionally, there is more data available for this time period, which lowers sample gaps and enhances the representativeness and accuracy of the analysis.

Important elements pertaining to the research question are reflected in the variables.

To ensure dependability and comparability, this study makes use of data that has been gathered and organised from international sources, such as the World Bank. The study uses a large dataset with 241 observations across seven important variables to investigate the connection between foreign direct investment (FDI) inflows and ease of doing business.

Both numerical and category variables are present in the dataset. Key elements of a country's economic and social growth are described by numerical variables, such as GDP per capita (in current US dollars), the percentage of the population that lives in urban areas, and foreign direct investment (as a percentage of GDP). Institutional and regulatory characteristics are represented by categorical variables, such as the political stability index and corruption control.

A statistical overview includes count, mean, median, standard deviation, minimum, and maximum values of variables.

Table 1. Variables description

Type of Variable	Name	Shorten Name	Units of measurement	Description
Dependent	Foreign direct investment, net inflows	fdi	current US\$	Foreign direct investment refers to direct investment equity flows in the reporting economy. It is the sum of equity capital, reinvestment of earnings, and other capital
Explanatory	Ease of doing business score	eodb	Score 0-100	The ease of doing business scores benchmark economies with respect to regulatory best practice, showing the proximity to the best regulatory performance on each Doing Business indicator

Control	GDP per capita	gdp	current US\$	GDP per capita is gross domestic product divided by midyear population. GDP is the sum of gross value added by all resident producers in the economy plus any product taxes and minus any subsidies not included in the value of the products
Control	Unemployment, total	unemployment	% of total labor force	Unemployment refers to the share of the labor force that is without work but available for and seeking employment.
Control	Urban population	urban	% of total population	Urban population refers to people living in urban areas as defined by national statistical offices. The indicator is calculated using World Bank population estimates and urban ratios from the United Nations World Urbanization Prospects.
Control	Political Stability and Absence of Violence/Terrorism	political	score	Political Stability and Absence of Violence/Terrorism measures perceptions of the likelihood of political instability and/or politically-motivated violence, including terrorism
Control	Control of Corruption	corruption	score	Control of Corruption captures perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as "capture" of the state by elites and private interests

Table 2. Data Summary

	N	Mean	Median	SD	Min	Max
fdi	171	9712166106.95	1024779237.91	35971385433.36	-127018529907.98	315984000000.00
eodb	171	64.37	65.97	13.04	20.04	86.76
gdp	171	14625.67	5807.84	20056.15	216.97	112726.44
unemployment	171	7.16	5.02	5.63	0.10	29.05
urban	171	59.21	60.04	22.78	13.25	100.00
political	171	-0.12	-0.08	0.90	-2.65	1.62
corruption	171	-0.07	-0.30	0.97	-1.72	2.13

FDI (fdi): The mean FDI is significantly high at approximately \$9.71 billion with a very large standard deviation of \$35.97 billion indicating high variability in the data points due to extreme values (minimum of approximately -\$127 billion and maximum of approximately \$316 billion). This suggests that while some countries attract substantial FDI, others may experience negative inflows.

EODB (eodb): The mean score is around 64.37, with a median of 65.97, indicating that most countries have relatively favorable business environments but with some outliers as shown by the minimum score of 20.04 and a maximum score of 86.76.

GDP (gdp): The average GDP is approximately \$14,625, with a median of \$5,807 suggesting that while some economies are very large (as indicated by the maximum GDP), many are significantly smaller.

Unemployment Rate(unemployment): The average unemployment rate is about 7.16%, with a minimum of 0.10% and a maximum of 29.05%, indicating considerable disparities in employment across different regions or countries.

Urban Population(urban): With an average urbanization rate of 59.21%, there is a significant range from 13.25% to 100%, reflecting varying levels of urban development.

Political Stability (political): The mean value of -0.12 indicates slight political instability on average across the dataset with a range from highly unstable (-2.65) to stable (1.62).

Control of Corruption(corruption): The average score for Control of Corruption is approximately -0.07, with a minimum score of -1.72 and a maximum score of 2.13. This indicates considerable disparities in perceptions of corruption across different countries.

Graphical Analysis

Scatterplots and histograms visualize relationships between variables like Ease of Doing Business (EoDB) and FDI inflows.

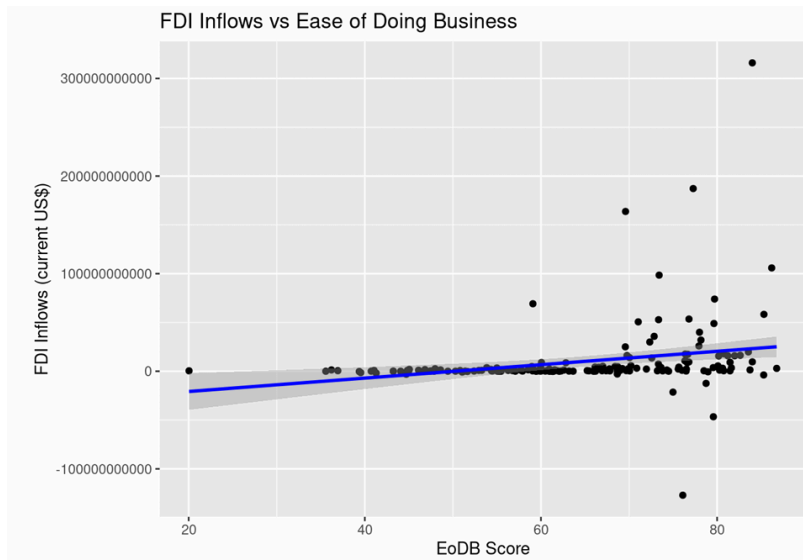


Figure 1. FDI & EoDB scatterplot

A positive correlation suggests that improving ease of doing business can be an effective strategy for countries seeking to enhance their attractiveness to foreign investors. However, there should be considered outliers, for developing comprehensive policies that consider all factors influencing FDI decisions.

Outliers are removed ("Luxembourg", "United States", "China", "Singapore", "Hungary", "Netherlands", "Ireland") because their extreme FDI values could distort regression analysis. For example, Luxembourg has exceptionally high FDI due to tax-related anomalies.

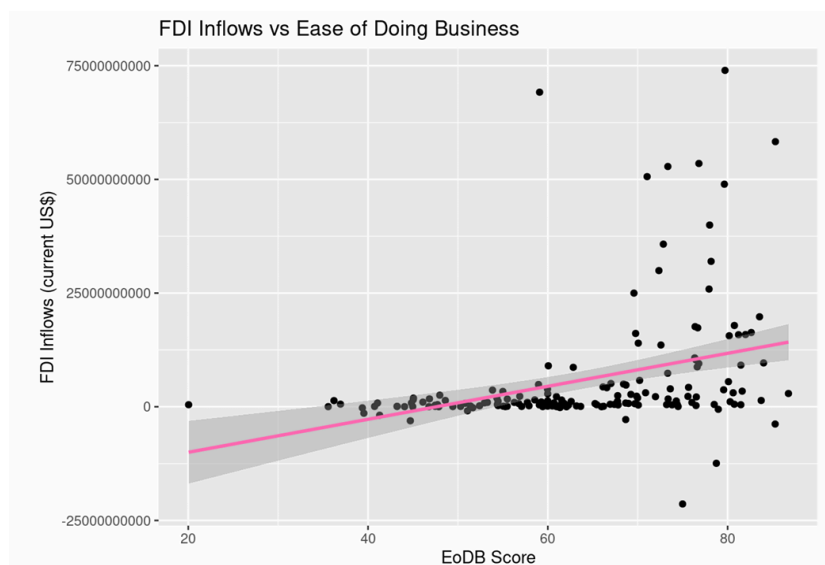


Figure 2. Cleared FDI & EoDB scatterplot

The removal of outliers affected the slope and provides a clearer understanding of the relationship between these two variables, free from the influence of extreme values that may distort the results. The positive correlation indicated by the linear trend line suggests that as EoDB scores increase—reflecting improved regulatory environments—FDI inflows also tend to increase proportionately.

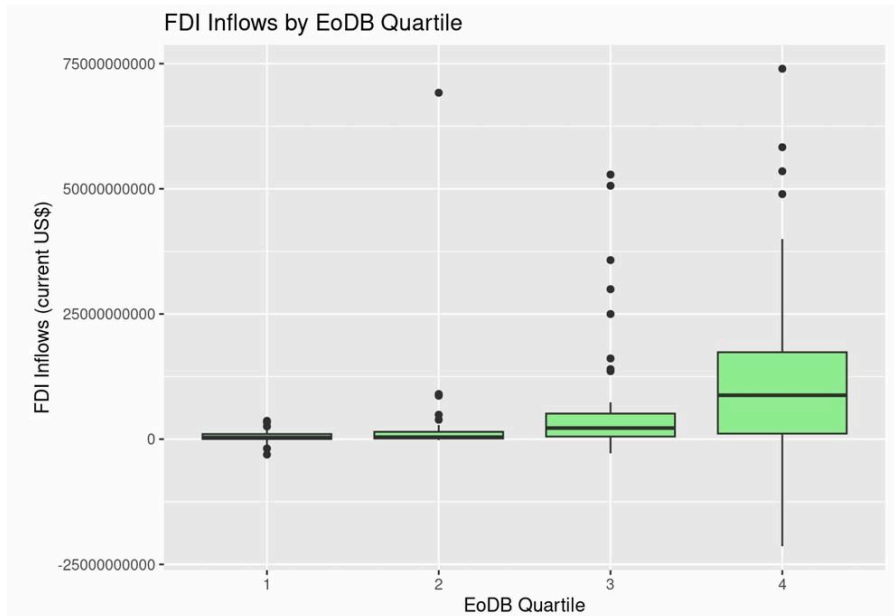


Figure 3. FDI Inflows by EoDB Quartile

Division by quartiles was made in order to compare groups: it helps to see how FDI inflows vary across countries with different levels of economic development. Boxplot outlines positive correlation: higher EoDB quartiles generally receive greater FDI inflows, suggesting that the level of the regulatory environment affects its attractiveness to foreign investors. It is extremely important to indicate unevenness: the outliers significantly vary within quartiles, especially among high EoDB countries.

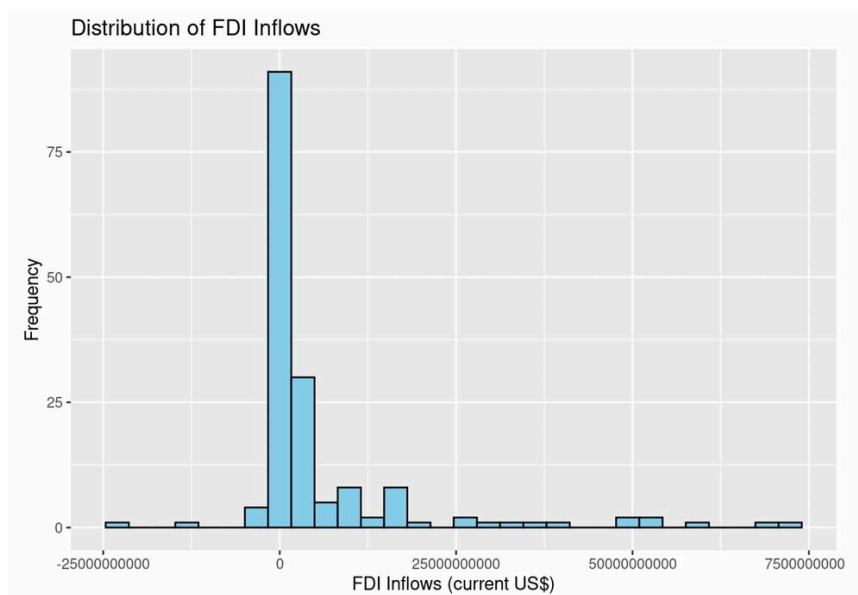


Figure 4. Distribution of FDI Inflows

The histogram was executed to detect skewness of data. The data turns out to be right-skewed. Furthermore, it shows that further transformation of data is needed to improve the performance of statistical analysis.

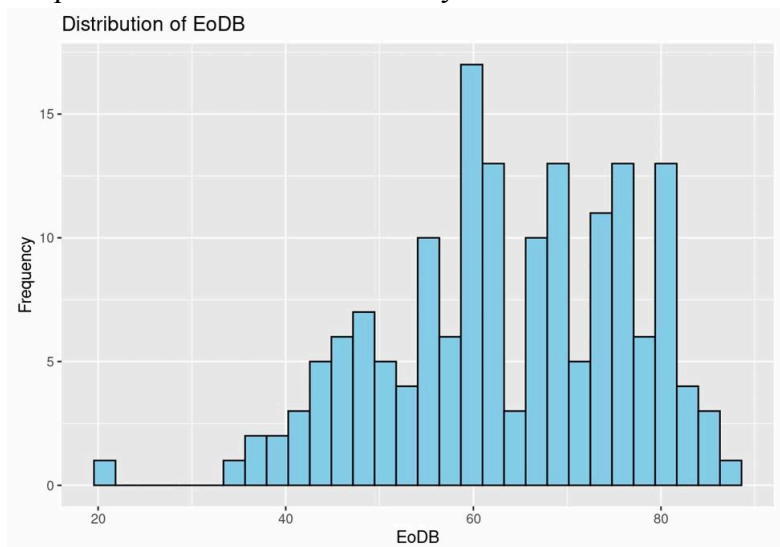


Figure 5. Distribution of EoDB

The histogram represents that the majority of countries have an EoDB index in the 40-80 range, suggesting a moderately business-friendly environment. However, a small number of countries face significant challenges, reflected in scores below 40. The two peaks hint at potential clusters based on shared regional or economic characteristics.

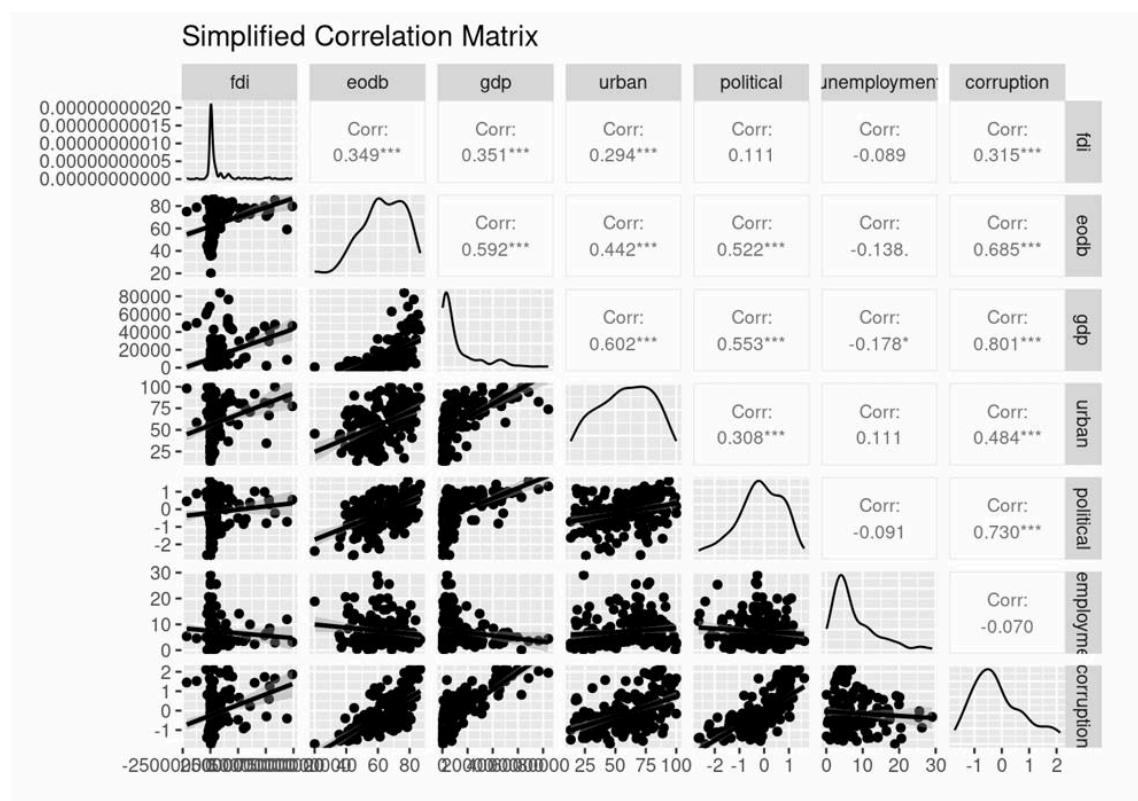


Figure 6. Correlation matrix

The correlation matrix represents the relationship between dependent, explanatory and control variables. It is important for control variables to be correlated with the dependent variable because this indicates that they may help explain some of the variance in the outcome. However, control variables should ideally not be highly correlated with explanatory variables to avoid multicollinearity issues.

Key Observations:

1. fdi and eodb (Corr: 0.349): A moderate positive relationship suggests that improving ease of doing business can be a key factor in attracting foreign direct investment.
2. eodb and gdp (Corr: 0.592): Strong correlation indicates that better business environments are associated with higher economic output, indirectly supporting FDI.
3. fdi and corruption (Corr: 0.315): Better control of corruption positively influences FDI, highlighting the importance of transparent institutional frameworks.
4. eodb and corruption (Corr: 0.685): Ease of doing business improves significantly in environments with better corruption control, reinforcing its role in creating investment-friendly conditions.
5. fdi and urban (Corr: 0.294): Urbanization moderately supports FDI, potentially through infrastructure and market access advantages.

Formulation and justification of models

Based on the provided code and analysis in R, here are 6 models outlined:

1. Simplified Model (Linear Regression)

Dependent variable: FDI_i - foreign direct investment.

Independent variable: $EoDB_i$ — an indicator of the ease of doing business.

Model: $fdi \sim eodb$

Mathematical form: $FDI_i = \beta_0 + \beta_1 * EoDB_i + \varepsilon_i$

2. Baseline Model

Dependent variable: FDI_i - foreign direct investment.

Independent variables:

$EoDB_i$ — an indicator of the ease of doing business;

GDP_i - Gross domestic product per capita;

$Unemployment_i$ — unemployment rate;

$Urban_i$ — the level of urbanization;

$Political_i$ — political stability;

$Corruption_i$ — the corruption control index.

Model: $fdi \sim eodb + gdp + unemployment + urban + political + corruption$

Mathematical form: $FDI_i = \beta_0 + \beta_1 * EoDB_i + \beta_2 * GDP_i + \beta_3 * Unemployment_i + \beta_4 * Urban_i + \beta_5 * Political_i + \beta_6 * Corruption_i + \varepsilon_i$

3. Log-Lin Model (Log(FDI) ~ EoDB)

Dependent variable: $\log(FDI_i)$ - the logarithm of foreign direct investment.

Independent variables:

$EoDB_i$ — an indicator of the ease of doing business;

GDP_i — Gross domestic product per capita;

$Unemployment_i$ — unemployment rate;

$Urban_i$ — the level of urbanization;

$Political_i$ — political stability;

$Corruption_i$ — the corruption control index.

Model: $\log(fdi) \sim eodb + gdp + unemployment + urban + political + corruption$

Mathematical form: $\log(FDI_i) = \beta_0 + \beta_1 * EoDB_i + \beta_2 * GDP_i + \beta_3 * Unemployment_i + \beta_4 * Urban_i + \beta_5 * Political_i + \beta_6 * Corruption_i + \varepsilon_i$

4. Log-Log Model (Log(FDI) ~ EoDB + Controls)

Dependent variable: $\log(FDI_i)$ - the logarithm of foreign direct investment.

Independent variables:

$\log(\text{EoDB}_i)$ — the logarithm of the ease of doing business indicator;

GDP_i — Gross domestic product per capita;

Unemployment_i — unemployment rate;

Urban_i — the level of urbanization;

Political_i — political stability;

Corruption_i — the corruption control index.

Model: $\log(\text{fdi}) \sim \log(\text{eodb}) + \text{gdp} + \text{unemployment} + \text{urban} + \text{political} + \text{corruption}$

Mathematical form: $\log(\text{FDI}_i) = \beta_0 + \beta_1 * \log(\text{EoDB}_i) + \beta_2 * \text{GDP}_i + \beta_3 * \text{Unemployment}_i + \beta_4 * \text{Urban}_i + \beta_5 * \text{Political}_i + \beta_6 * \text{Corruption}_i + \varepsilon_i$

5. Model (Log(FDI) ~ EoDB × Political)

Dependent variable: $\log(\text{FDI}_i)$ - the logarithm of foreign direct investment.

Independent variables:

EoDB_i — ease of doing business indicator;

Political_i — political stability;

$\text{EoDB}_i * \text{Political}_i$ — the interaction of ease of doing business and political stability

Model: $\log(\text{fdi}) \sim \text{eodb} + \text{political} + \text{eodb} : \text{political}$

Mathematical form: $\log(\text{FDI}_i) = \beta_0 + \beta_1 * \text{EoDB}_i + \beta_2 * \text{Political}_i + \beta_3 * (\text{EoDB}_i * \text{Political}_i) + \varepsilon_i$

This model examines whether the impact of Ease of Doing Business (EoDB) on FDI inflows depends on political stability. Political stability influences investor confidence; in politically stable countries, the positive effects of EoDB are amplified, as stable environments reduce risk and increase the credibility of reforms. Conversely, in unstable settings, high EoDB scores may fail to attract FDI due to unpredictability

6. Model Log(FDI) ~ EoDB × Corruption

Dependent variable: $\log(\text{FDI}_i)$ - the logarithm of foreign direct investment.

Independent variables:

EoDB_i — ease of doing business indicator;

Corruption_i — the corruption control index;

$\text{EoDB}_i * \text{Corruption}_i$ — the interaction of the ease of doing business and the level of corruption control.

Model: $\log(\text{fdi}) \sim \text{eodb} + \text{corruption} + \text{eodb} : \text{corruption}$

Mathematical form: $\log(\text{FDI}_i) = \beta_0 + \beta_1 * \text{EoDB}_i + \beta_2 * \text{Corruption}_i + \beta_3 * (\text{EoDB}_i * \text{Corruption}_i) + \varepsilon_i$

This model investigates how the relationship between EoDB and FDI inflows changes with corruption control. Corruption undermines the benefits of EoDB by increasing costs and uncertainty. Strong corruption control enhances EoDB's effect on FDI by ensuring transparent and efficient systems, making the business environment more attractive to investors

Purpose of Interaction Models

Interaction models reveal how EoDB's effect on FDI is conditioned by governance factors like political stability and corruption control, providing nuanced insights for policymaking. They show that EoDB reforms alone may not suffice; complementary improvements in governance are often necessary to maximize FDI inflows

To prevent distortions in the evaluation of the primary link between the independent and dependent variables, control variables are incorporated into the models to take into consideration variables that could have an impact on the dependent variable.

1) Taking out the hidden bias variable

Control variables prevent results from being distorted by removing the influence of variables that correlate with X and affect Y. Example: The impact of ease of doing business on foreign direct investment (FDI) may be exaggerated if, for example, political stability is not taken into consideration.

2) Making the primary relationship clear

They enable us to minimise the impact of unimportant elements while emphasising the importance of a crucial variable. For instance, taking urbanisation into account ignores how population density affects the connection between investments and the economic climate.

3) Increasing the model's accuracy

Control variables increase the model's explanatory power by accounting for additional factors that influence Y. For instance, some of the non-X fluctuations in FDI can be explained by adding GDP per capita and unemployment.

4) Lowering the bias variable that is missing

The estimates are biased due to the lack of important components. Control variables assist in preventing this. For instance, corruption can have a significant impact on FDI. Estimates of other variables will be inaccurate if it is not taken into consideration.

The use of the natural logarithm (log) in modeling is often associated with several advantages:

1) Interpreting coefficients linearly

In the event that both the independent and dependent variables are logarithmic: When EoDB is changed by 1%, the coefficients turn into elasticities (e.g., the change in FDI as a percentage). This makes it easier to understand the findings, particularly in economic models where proportionate changes are relevant. Assuming that $\log(\text{FDI}_i) = \beta_0 + \beta_1 * \log(\text{EoDB}_i)$, then β_1 displays the percentage change in FDI as a result of a 1% rise in EoDB.

2) Variance stabilisation (homoscedasticity)

The issue of heteroscedasticity frequently affects data with significant variability or disparate scales. By reducing scale disparities, the logarithm improves the model estimation's accuracy and creates a more consistent residual distribution. For instance, in the absence of the logarithm, the variance is FDI, which can increase as GDP per capita increases. This reliance is lessened when a logarithm is used.

3) Diminishing the influence of emissions

Economic indicators like GDP per capita and FDI frequently have extreme values (emissions). Since the logarithmic function grows more slowly than the linear one, logarithm lowers their relative weight. The difference is significant for $FDI = 1,000$ and $FDI = 10$, but following the logarithm, the model is less distorted and the difference is smaller.

4) Proportional change modelling

Rather than using absolute values, economic variables are frequently examined in terms of their proportionate changes. We can simplify the process and model proportionality relations with the logarithm. As an illustration, the difference between $FDI = 100$ and $FDI = 200$ is twice as big as the difference between $FDI = 500$ and $FDI = 1,000$ (also a two-fold increase). This is reflected in the model's logarithm

5) Complex relationships' linearity

Using a logarithm can make many intricate nonlinear interactions between variables linear. This makes it easier to analyse and interpret the model.

Due to the fact that the natural logarithm can only be taken from a positive number, some of the FDI data is zeroed out during the regression analysis; however, this convention of the model can be respected due to the fact that countries with negative FDI flows represent either African countries with very low investment attractiveness or countries with a huge number of outward investment flows due to active offshoring, which is irrelevant for our model

Expected results

The results are expected to show the following:

1. Significant impact of the business environment on investment attraction: Countries with high levels of ease of doing business are expected to show more FDI due to simplified administrative procedures and lower regulatory costs for investors.
2. Importance of additional factors: GDP per capita, urbanization, low levels of unemployment and corruption, and political stability are likely to strengthen or weaken the impact of ease of doing business on FDI, demonstrating their contribution to creating an investment-attractive environment.
3. Identification of key factors in attracting FDI: The study is expected to highlight the key elements that most strongly influence the volume of foreign investment, helping to identify areas for improving the investment climate.
4. Practical relevance of the results: The study may identify specific aspects of the investment climate that governments can improve to enhance their countries' competitiveness in the international investment market.

The results that will be obtained after the study will help to better understand the nature of the relationship between economic and demographic conditions and investments, as well as develop recommendations for the formation of effective measures to improve the investment climate.

Regression results analysis

- 1) In the simplified regression model, FDI was taken as the dependent variable and Ease of doing business was taken as the regressor, excluding control variables. According to the regression results, we can say that even with the simplified model, the main regressor turns out to be statistically significant at the significance level of 0.01 and when Ease of Doing Business changes by 1 point, FDI inflows change by \$686,404,628.5. However, for all this, the explanatory power of the model is minimal due to the extremely low value of adj. R²
- 2) In the second model, we took into account all control variables, keeping the dependent variable the same. However, in this case, no variable showed statistical significance. Moreover, the FDI became less dependent on the main regressor, and the model itself is still statistically irrelevant
- 3) In the third model, we normalize the dependent variable by the logarithm and leave the regressor and controls unchanged. As can be seen, all variables are significant at least at the 5% level of significance: Political stability and unemployment rate at the 0.1% level of significance, and the main regressor, GDP per capita and urbanization rate at the 0.01% level. Moreover, the explanatory power of the model can be considered strong at adj. R² = 0,54
- 4) Next, we decided to test the hypothesis of the effect of proportional changes in Ease of Doing Business, so we took Log(EoDB) as the main regressor, but we obtained only small reductions in the statistical significance of each of the variables, judged by P-value, and of the model itself by adj. R², folding in the conclusion that this is definitely not model strengthening
- 5) The last two models were formulated to test another hypothesis about the interactive effect of the main regressor influence in pairs with the least significant control variables from the main model. As can be seen, the statistical significance of Political Stability did not change in the model, but the interactive effect became positively influential in contrast to the main model with a negative influence of Political Stability: when the aggregate of EoDB and Political Stability grows by 1 point, FDI grows by 2.98%. At the same time, the interactive effect of the regressor and Corruption Control showed a statistically strong effect on EoDB at 0.01% significance level: when the aggregate of EoDB and Corruption Control increases, FDI increases by 4.01%. However, these results showed that the explanatory power of the models is significantly lower than that of the main model

An answer to a meaningful question within the framework of the analysis

The Ease of Doing Business (EoDB) has a significant impact on the volume of foreign direct investments (FDI) that countries attract. The analysis shows that high EoDB scores are directly linked to an increase in FDI inflows.

Ease of doing business encourages foreign investment by simplifying administrative procedures, reducing costs, and enhancing transparency. In countries with a favorable business climate, investors feel more confident, which increases capital inflows.

However, the effectiveness of EoDB is amplified in the presence of political stability and corruption control. These factors create an institutional environment that supports long-term investments. It is also important to consider the interaction of EoDB with economic and social parameters, such as urbanization, income levels, and employment. Together, these factors shape the overall investment climate, determining a country's attractiveness.

To increase the volume of foreign direct investment, countries need to focus not only on improving EoDB scores but also on strengthening institutional stability. This includes optimizing administrative processes, ensuring the rule of law, and reducing corruption. Such a comprehensive approach will help countries strengthen their position in the global capital market.

Critical analysis of the results obtained

The study demonstrated a number of limitations that should be considered when interpreting the results. First and foremost, the model simplifies economic reality, not capturing the specific characteristics of individual sectors. For instance, the technology and infrastructure sectors depend on the quality of institutions, whereas resource-based industries focus on the availability of natural resources. The model also excludes non-factorizable aspects, such as investor perceptions, cultural proximity, and geopolitical factors, which are difficult to quantify but play a significant role in investment decisions. Using data from only 2019 does not account for the dynamics of trends or the effects of global crises, such as the COVID-19 pandemic, which limits the ability to analyze the evolution of factors influencing FDI.

Furthermore, potential omissions were identified. The absence of factors such as exchange rate stability, openness of foreign trade, and infrastructure quality in the model may lead to distorted results. Regional characteristics are also outside the scope of the study, obscuring differences between countries with varying levels of development. For example, the determinants of FDI in developed economies may differ significantly from the factors influencing investment inflows in developing countries.

An interesting conclusion from the study is that, based on our models, political stability and control of corruption have a negative effect on FDI, which contradicts theoretical expectations. In the real world, it is expected that a country with lower levels of corruption would be more attractive to investments. Possible reasons for this can include methodological aspects: the sample may have included countries with authoritarian regimes, where political stability is achieved through centralization of power, reducing the effectiveness of corruption control. Cultural and historical factors may also play a role, as corruption in some countries remains systemic even in stable political environments. Finally, multicollinearity between political stability, corruption control, and other variables could have distorted the model's coefficients.

Analysis of the Limitations of the Study

1. Limited Data Scope: The research is predicated on 2019 data, which only provides a glimpse of the political and economic climate of that year. Long-term patterns and the changing nature of corporate environments might not be adequately captured by this. For example, the COVID-19 epidemic, which started in 2020, had a big impact on economic conditions around the world, and our analysis might not have taken into account how it influenced FDI. The way these elements change over time could be better understood with a more thorough, longitudinal approach.

2. Key Variables Ignored: Although the study took into account urbanisation, political stability, corruption control, unemployment and GDP per capita, other variables that can affect FDI were overlooked. These include the quality of the infrastructure, the availability of qualified labour, taxation policies, and the legal framework—all of which are essential for drawing in foreign investment. The study may not be able to present a completely accurate picture of the factors influencing FDI if these elements are excluded.

3. Data Reliability and Consistency: The analysis uses publicly accessible measures of urbanisation, GDP per capita, political stability, and ease of doing business. These indexes, however, may have inconsistencies because they were created using disparate approaches. Rankings of political stability, for instance, may be arbitrary, and variations in how nations present economic statistics may compromise the accuracy of the findings. The accuracy of the study might be increased by taking a more detailed approach to assessing these factors.

4. Causality Issues: While highlighting links between FDI and ease of doing business, the study does not prove causation. It is challenging to ascertain whether enhancing the ease of doing business would directly result in greater foreign investment or if other external factors influence these outcomes, even though a favourable business environment is linked to increased FDI. More sophisticated analytical methods, like randomised control trials or instrumental variable methods, could be used to prove causation.

Discussion part with suggestions for possible expansion of the study

Research perspectives must be expanded to include other aspects that can have a substantial impact on these processes in order to gain a greater knowledge of how variations in the ease of doing business affect the amount of foreign direct investment. Even though the research relies heavily on listed control variables, it's crucial to take into account other factors that could alter the type or strength of these variables' effects on luring in foreign investment. Given this, we may propose a number of avenues for the study's future advancement that will enable us to more precisely evaluate the influence of both internal and external factors on a nation's appeal to international investors.

One of these areas is to expand the time horizon of the study, including data not only for 2019, but also for several previous years, as well as for the future. This will help identify long-term trends and dependencies between changes in business conditions and FDI volumes. It is also worth conducting an in-depth comparative analysis by region, for example, Africa, Asia or Latin America, in order to understand how regional characteristics and local economic conditions can affect the attraction of foreign investment.

Since every economic sector may respond differently to changes in the business environment, it will be equally vital to examine how different industries recruit FDI. A more thorough examination of political stability, taking into account elements like the standard of governance, the legal system, and the battle against corruption, can also be beneficial. We can better understand the dynamics of shifts in a nation's appeal to foreign investors if we include an examination of social infrastructure, such as access to healthcare and education, as well as an examination of external economic factors, such as trade wars or global financial crises.

The influence of future changes in determinants on the amount of FDI can be more precisely predicted with the use of qualitative data, such as investor interviews, and alternative scenario modelling. This will give the study more real-world relevance and enable us to formulate more precise suggestions for policymakers and entrepreneurs looking to improve the climate for foreign investors.

The model can incorporate variables like tax laws, infrastructure quality, skilled labour availability, and currency rate stability. This will enable us to get a more comprehensive understanding of the elements that either encourage or inhibit the influx of foreign investment.

Finding effective methods for changing the company environment will be made easier by adding qualitative case studies to quantitative analysis. A study of nations where FDI has increased and EoDB ratings have significantly improved, for instance, can offer useful reform suggestions.

Taking into account the effects of technical developments, such the use of blockchain, fintech, and artificial intelligence, may help explain how digital transformation changes the business landscape and draws foreign direct investment. For example, countries that

incorporate technology into their regulatory procedures—such as blockchain for contract enforcement or automated licensing systems—may make their environments more appealing to international investors.

Future research could assess how a nation's environmental laws, green energy infrastructure, and ESG (Environmental, Social, Governance) policies affect FDI decisions as sustainability becomes a major concern for international investors. Greenfield investments in industries like sustainable agriculture and renewable energy may rise in nations that prioritise sustainable development.

Examining the relationship between a nation's ease of doing business rankings and its capacity to retain or draw foreign direct investment (FDI) during downturns as well as its resilience to economic shocks (such as pandemics and financial crises) could yield important insights. The recovery paths of countries with high and low EoDB rankings could be compared in comparative studies.

Conclusion

Our study has made a significant contribution to the development of understanding of the relationship between the financial and general economic attractiveness of countries and the foreign direct investment attracted. It shows the subtle effects of Ease of Doing Business on the volume of inward FDI. Our methodology includes not only the basic regression model, but also our formulated hypotheses of influence and data transformation, taking into account the mutual influence of control variables and the main regressor, which ultimately provides insight into the reliability of the model and the study itself. Nevertheless, some limitations of the study need to be taken into account, such as not taking into account the dynamics of the situation in the cross-sectional data, other variables that directly or indirectly affect FDI inflows, different ways of accounting and data collection by the official source and confirmation bias, which may have confused us while searching for the true causes of FDI changes across countries. The results we obtained can and should be used in further research works on this and related topics, taking into account other variables, factors and years. The study emphasizes the need for further research on this topic to build a comprehensive picture of all factors and aspects affecting the investment attractiveness of countries

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Appendix

Descriptive statistics graphs

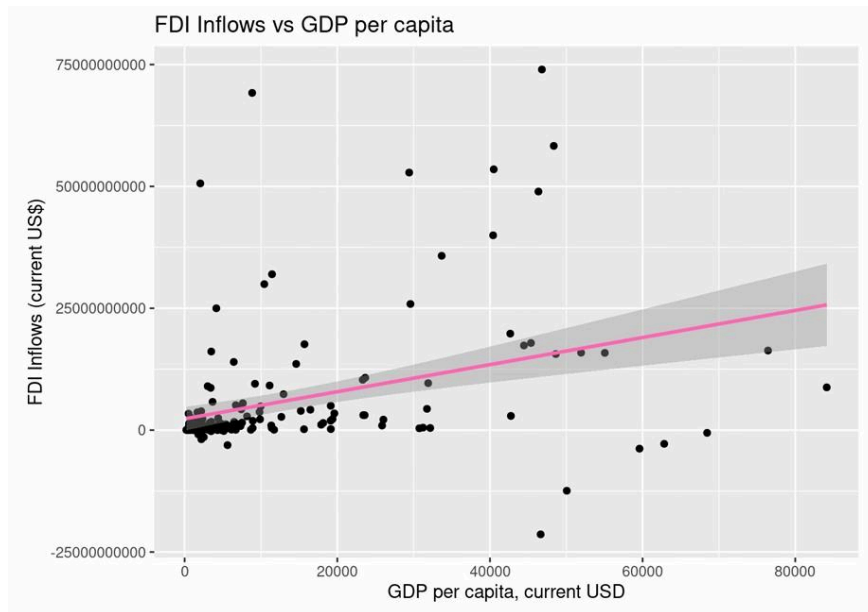


Figure 7. FDI & GDP per capita scatterplot

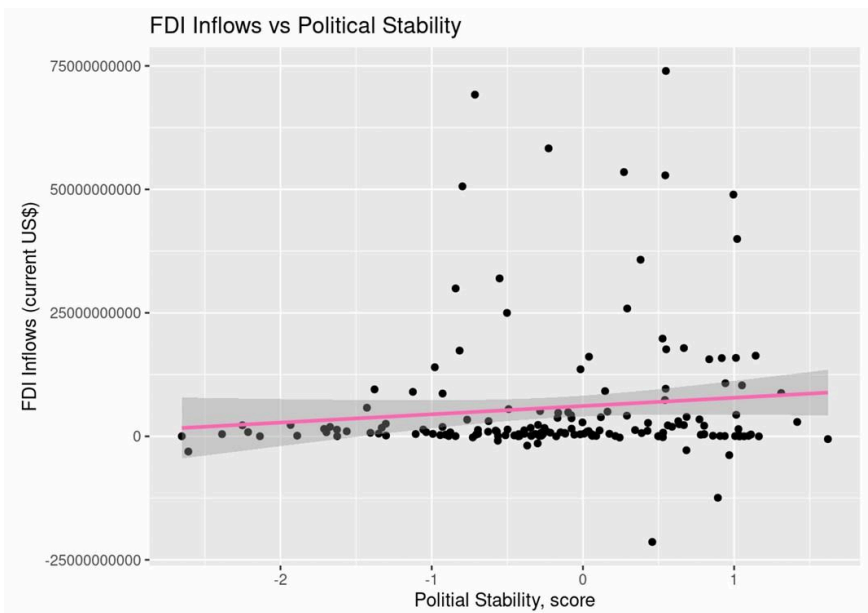


Figure 8. FDI & Political stability scatterplot

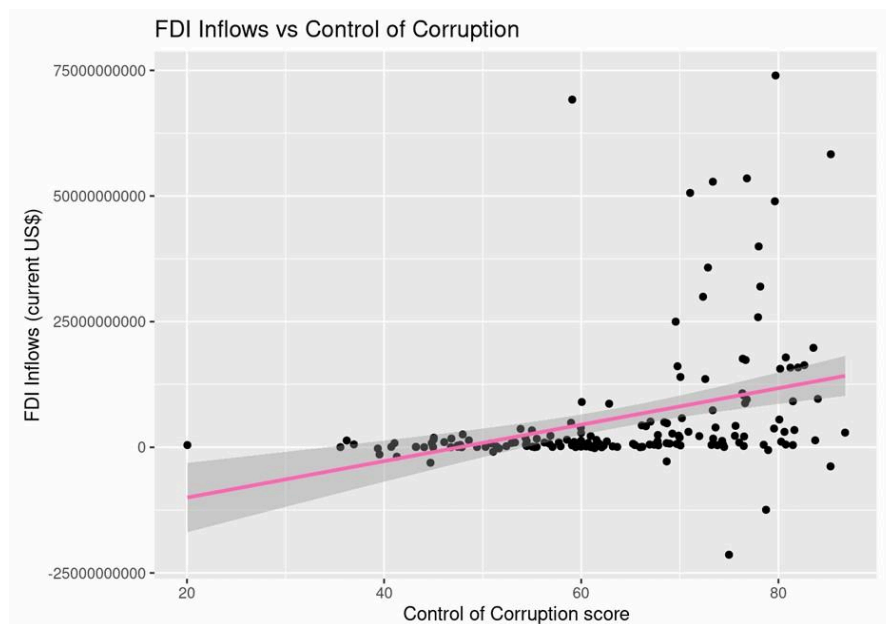


Figure 9. FDI & Corruption control scatterplot

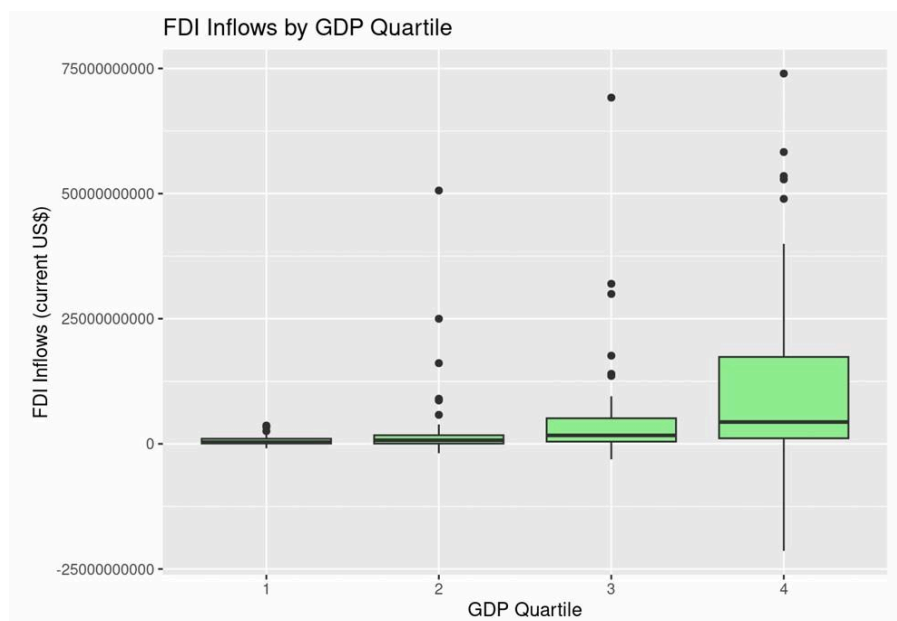


Figure 10. FDI Inflows by GDP Quartile

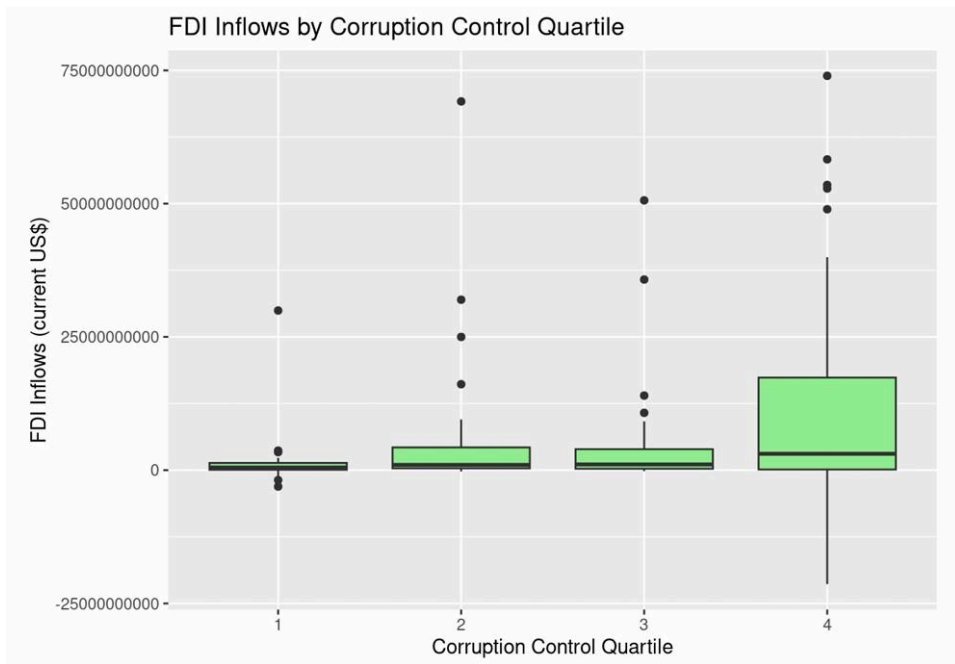


Figure 11. FDI Inflows by Corruption control Quartile

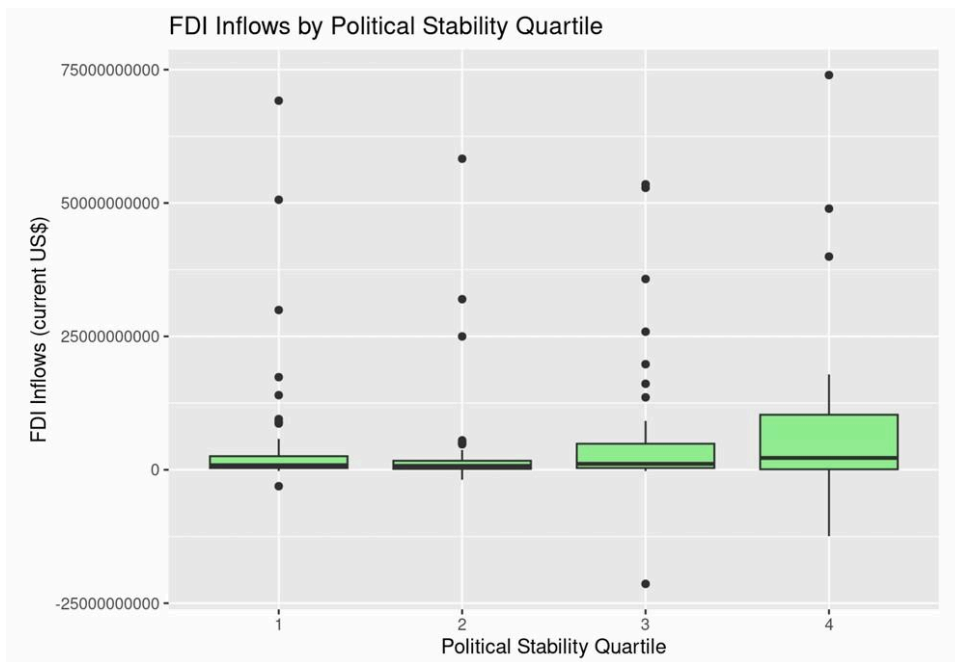


Figure 12. FDI Inflows by Political stability Quartile

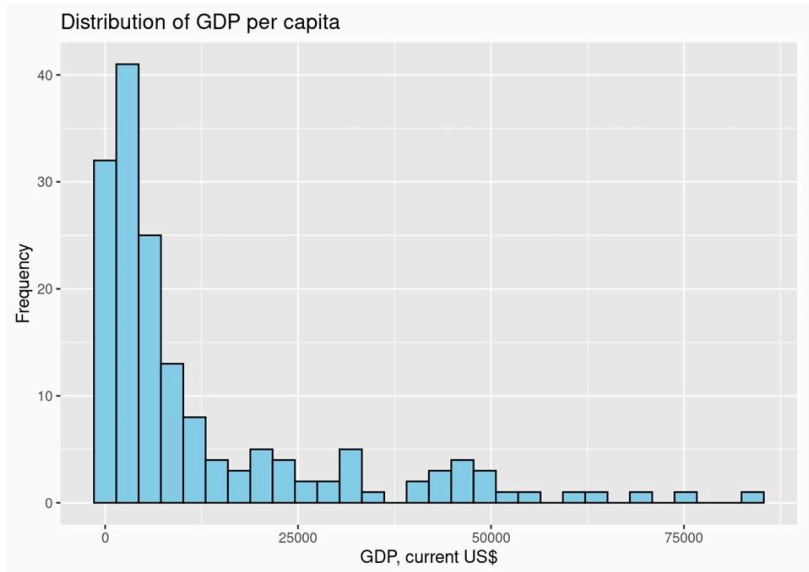


Figure 13. Distribution of GDP per capita

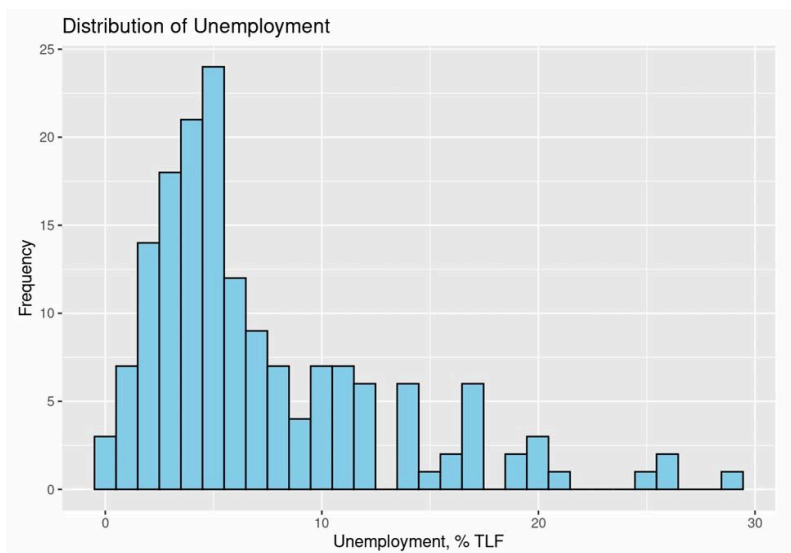


Figure 14. Distribution of Unemployment rate

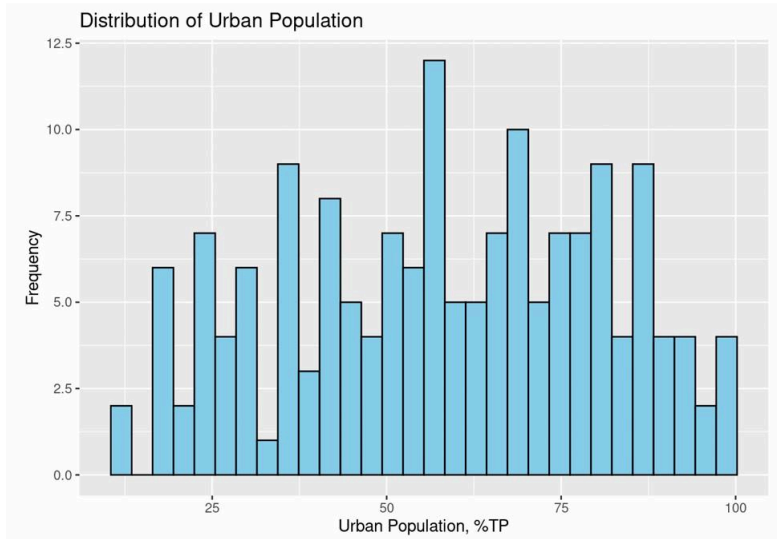


Figure 15. Distribution of Urbanization rate

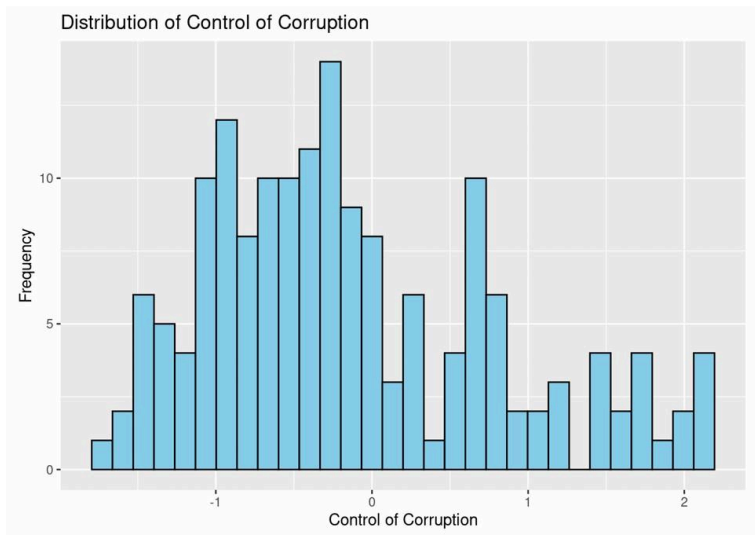


Figure 16. Distribution of Corruption control

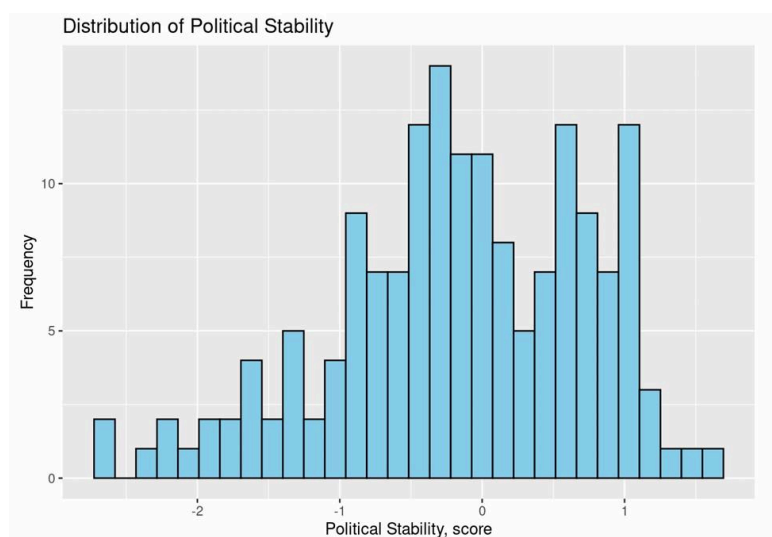


Figure 17. Distribution of Political stability and absence of terrorism

Table 3. Regression results

Model	Simple	Basic	Log-lin	Log-log	Inter.1	Inter.2
Dependent Var.	fdi	fdi	log_fdi	log_fdi	log_fdi	log_fdi
eodb	686,404,628.5** (250,786,970.4)	536,030,990.0. (280,719,135.6)	0.0775*** (0.0132)		0.1221*** (0.0135)	0.1168*** (0.0171)
gdp		806,195.0 (539,721.2)	5.08e-5*** (1.01e-5)	5.38e-5*** (1.05e-5)		
unemployment		-4,804,530.3 (271,348,672.2)	-0.0636** (0.0222)	-0.0638** (0.0228)		
urban		-25,345,998.4 (138,613,167.8)	0.0271*** (0.0069)	0.0282*** (0.0069)		
political		--2,147,483,648.1 (-2,147,483,648.1)	-0.5754** (0.1986)	-0.6204** (0.2034)	-2.160** (0.7170)	

corruption		--2,147,483,648.9 (-2,147,483,648.6)	-0.4562. (0.2652)	-0.3945 (0.2672)		-2.737** (0.8295)
log_eodb				4.071*** (0.9266)		
eodb x political					0.0298** (0.0108)	
eodb x corruption						0.0401*** (0.0106)
Adj. R2	0.05635	0.10888	0.54074	0.52859	0.36803	0.37928

Regression models graphs analysis

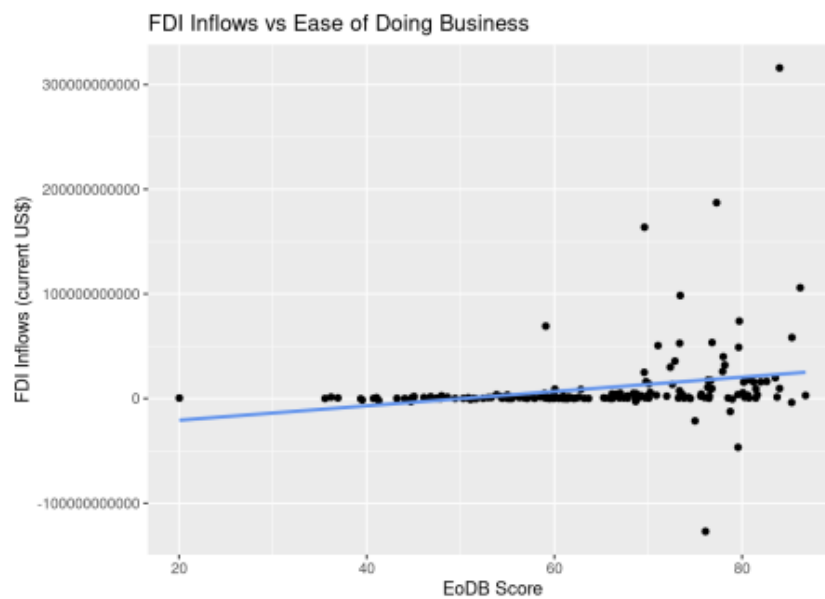


Figure 18. FDI & EoDB regression scatterplot

The regression line of the Simplified Model shows a weak positive impact of the Ease of Doing Business indicator on FDI inflows. The points are highly scattered, especially at high EoDB Score values. This suggests that the relationship between these variables is weak and not unambiguous

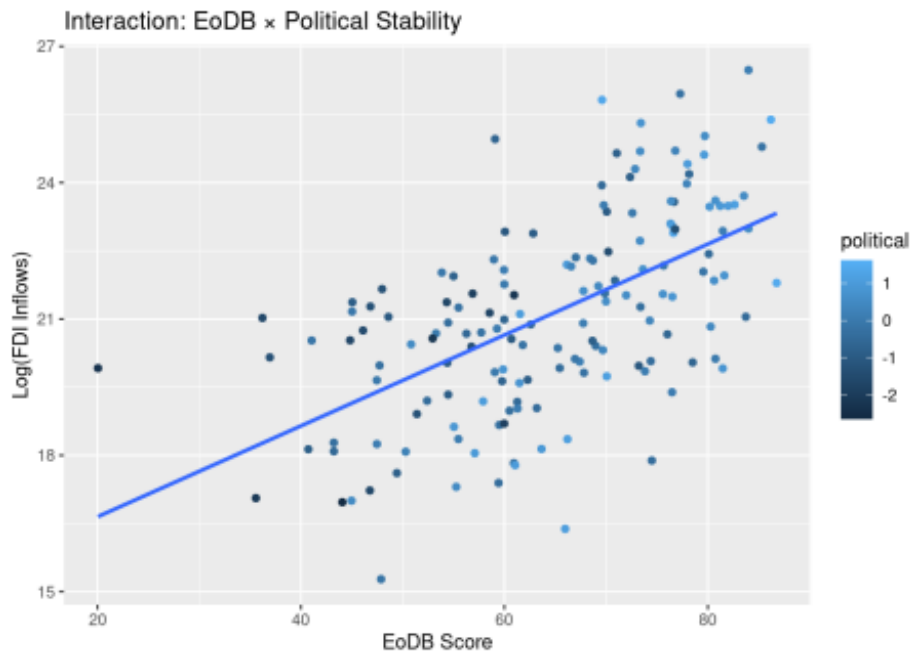


Figure 19. Interaction model by Political stability scatterplot

The regression line shows a clear positive impact of EoDB Score on the approximated Log(FDI Inflows) indicator. Political stability amplifies the positive impact of the EoDB on investment inflows. In countries with high stability, improvements in the business environment lead to significantly higher investments

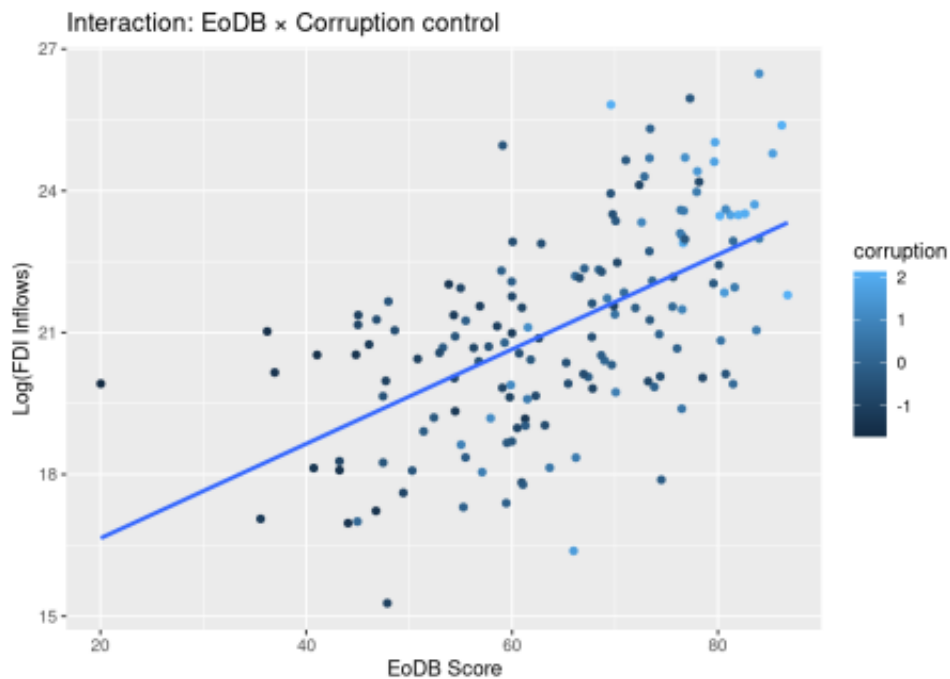


Figure 20. Interaction model by Corruption control scatterplot

When corruption control is high (light dots), Log(FDI) values are higher compared to low control. When corruption control is low (dark points), the dependence weakens and even high EoDB values do not lead to such significant investments